

EPaxos Log Metrics: Column Reference

(Per-Replica and Cluster-Wide)

General Notes

- **Starts** refer only to instances that a given replica *initiated* (proposal start seen in its own log).
- **Fast vs. Slow classification:** if both fast- and slow-path signals appear for the same instance, the outcome is classified as *slow* (fallback overrides an early fast signal).
- All “%” columns are percentages (0–100), not fractions.
- The *shared key* is the configured hot key (e.g., `share_key`); *unique* keys are any others.

Per-Replica Metrics (columns in `replica_metrics.csv`)

Column	Description
<code>replica_id</code>	Numeric id of the replica whose log was analyzed.
<code>log_file</code>	File name the metrics were computed from.
<code>starts</code>	Count of instances started (proposal initiated) on this replica. Denominator for most rates.
<code>fast</code>	Number of started instances that completed via the fast path.
<code>slow</code>	Number of started instances that completed via the slow path (includes cases that first showed a fast signal then fell back).
<code>unknown_outcome</code>	Started instances for which neither fast nor slow completion was observed in the log.
<code>fast_rate_pct</code>	Percentage of starts that were fast: $100 \times \frac{\text{fast}}{\text{starts}}$.
<code>slow_rate_pct</code>	Percentage of starts that were slow: $100 \times \frac{\text{slow}}{\text{starts}}$.
<code>shared_starts</code>	Number of starts whose command key equals the configured shared (hot) key.
<code>unique_starts</code>	Number of starts whose command key is not the shared key.
<code>unknown_key_starts</code>	Number of starts for which the key could not be parsed from the log line.
<code>shared_key_fraction_pct</code>	Percentage of starts that used the shared key: $100 \times \frac{\text{shared_starts}}{\text{starts}}$.
<code>slow_on_shared</code>	Count of shared-key starts that ended slow.
<code>slow_on_unique</code>	Count of unique-key starts that ended slow.
<code>slow_rate_shared_only_pct</code>	Slow rate among shared-key starts only: $100 \times \frac{\text{slow_on_shared}}{\text{shared_starts}}$ (0% if denominator is 0).
<code>slow_rate_unique_only_pct</code>	Slow rate among unique-key starts only: $100 \times \frac{\text{slow_on_unique}}{\text{unique_starts}}$ (0% if denominator is 0).
<code>conflicted_instances</code>	Number of started instances that logged at least one dependency/conflict with another instance while building deps.
<code>conflict_edges</code>	Total number of conflict edges observed across started instances (can exceed <code>conflicted_instances</code> if one instance conflicts with multiple others).
<code>conflict_rate_pct</code>	Percentage of starts that saw at least one conflict: $100 \times \frac{\text{conflicted_instances}}{\text{starts}}$.

Cluster-Wide Metrics (columns in `cluster_metrics.csv`)

Interpretation tips. A healthy EPaxos run typically shows (i) high `fast_rate_pct` for unique keys, (ii) elevated `slow_rate_shared_only_pct` when the shared (hot) key is exercised concurrently, and (iii) consistent totals across replicas with low `unknown_outcome`.

Column	Description
<code>starts</code>	Sum of <i>per-replica</i> <code>starts</code> (no double-counting; each instance counted on its initiating replica only).
<code>fast</code>	Sum of per-replica <code>fast</code> .
<code>slow</code>	Sum of per-replica <code>slow</code> .
<code>unknown_outcome</code>	Sum of per-replica <code>unknown_outcome</code> .
<code>fast_rate_pct</code>	Cluster fast percentage: $100 \times \frac{\text{fast}}{\text{starts}}$.
<code>slow_rate_pct</code>	Cluster slow percentage: $100 \times \frac{\text{slow}}{\text{starts}}$.
<code>shared_starts</code>	Sum of per-replica <code>shared_starts</code> .
<code>unique_starts</code>	Sum of per-replica <code>unique_starts</code> .
<code>unknown_key_starts</code>	Sum of per-replica <code>unknown_key_starts</code> .
<code>shared_key_fraction_pct</code>	Cluster fraction of starts using the shared key: $100 \times \frac{\text{shared_starts}}{\text{starts}}$.
<code>slow_on_shared</code>	Sum of per-replica <code>slow_on_shared</code> .
<code>slow_on_unique</code>	Sum of per-replica <code>slow_on_unique</code> .
<code>slow_rate_shared_only_pct</code>	Cluster slow rate among shared-key starts: $100 \times \frac{\text{slow_on_shared}}{\text{shared_starts}}$ (0% if denominator is 0).
<code>slow_rate_unique_only_pct</code>	Cluster slow rate among unique-key starts: $100 \times \frac{\text{slow_on_unique}}{\text{unique_starts}}$ (0% if denominator is 0).
<code>conflicted_instances</code>	Sum of per-replica <code>conflicted_instances</code> .
<code>conflict_edges</code>	Sum of per-replica <code>conflict_edges</code> .
<code>conflict_rate_pct</code>	Cluster conflict percentage: $100 \times \frac{\text{conflicted_instances}}{\text{starts}}$.